┌Def.

Let G be a group and A be a non-empty subset. A map

$$\alpha : G \times A \to A : (g, a) \mapsto g.a$$

is called a group action of G on A if it satisfies

1) For all $a \in A$, $1_G \cdot A = A$
2) For all $g_1, g_2 \in A$, and $a \in A$, $(g_1 g_2) \cdot a = g_1 \cdot (g_2 \cdot a)$

┌Prop. Given a group action α of G on A, there exists a homomorphism

$$\varphi_\alpha : G \to S_A : g \mapsto \sigma_g \quad \text{where} \quad \sigma_g : A \to A : a \mapsto ga.$$

which is called the permutation representation associated to the given action.
Conversely, given a homomorphism $\varphi : G \to S_A$, there exists a group action

$$\alpha_\varphi : G \times A \to A : (g, a) \mapsto \varphi(g)(a)$$

Proof. We only need to check that the given map is well-defined,
Let $g \in G$ be fixed. Then, $\sigma_g \circ \sigma_{g^{-1}}(a) = \sigma_{g^{-1}} \circ \sigma_g(a) = a$, therefore $\sigma_g \in S_A$.

Thm.

Let G be a group and $H \leq G$ be a subgroup.
Consider G acts on $\{gH \mid g \in G\}$ by left multiplication, and affored homomomorphism

$$\pi_H : G \to S_A : g \mapsto \sigma_g \quad \text{where } A = \{gH \mid g \in G\}$$
$$\text{and } \sigma_g(xH) = gxH.$$

Then G acts transitively on A and the stabilizer $G_H = H$, and

$$\ker \pi_H = \bigcap_{x \in G} xHx^{-1} \leq H$$

is the largest normal subgroup of G contained in H.

Proof. Given $aH, bH \in A$, $bH = ba^{-1}(aH)$, so aH and bH lie in same orbit.
And, since $gH = H \iff g \in H$,

$$G_H = \{g \in G \mid gH = H\} = H.$$

To show the last assertion, by definition,

$$\ker \pi_H = \bigcap_{x \in G} G_{xH}$$

and $G_{xH} = \{g \in G \mid gxH = xH\} = \{g \in G \mid x^{-1}gx \in H\} = xHx^{-1}$.
Moreover, $\ker \pi_H \trianglelefteq G$ because it is kernel of the homomomorphism, and for any normal subgroup $N \trianglelefteq G$ of G contained in H,

for all $x \in G$, $N = xNx^{-1} \leq xHx^{-1}$

therefore $N \leq \ker \pi_H$. ☒

Note: given a subgroup $H \leq G$, $G/\ker \pi_H \cong \operatorname{Im} \pi_H$.

「Thm」

Let G be a finite group of order n, and let p be the smallest prime dividing n. Then, every subgroup of index p is normal.

Proof. Let $H \leq G$ with $|G:H| = p$. Let G acts on left cosets of H and

$$\pi_H : G \to S_p$$

be the afforded permutation of the given action.
Then, $\ker \pi_H \leq H$, let $K = \ker \pi_H$ and $k = |H:K|$. Then,

$$|G:K| = |G:H| \cdot |H:K| = pk.$$

Since $G/K \cong \mathrm{Im}\, \pi_H \leq S_p$, so pk divides $p!$ by the Lagrange's Theorem. Therefore, k divides $(p-1)!$. But all prime factors of $(p-1)!$ are less than p, by the minimality of p, k must be 1, that is, $H = K = \ker \pi_H \trianglelefteq G$ ☒

Note: let G acts on finite set A which has m elements.
Then, the afforded permutation $\varphi : G \to S_m$ satisfies:

$$G/\ker \varphi \cong \mathrm{Im}\, \varphi \leq S_m \text{ implies } |\mathrm{Im}\, \varphi| \,\Big|\, |G| \text{ and } |\mathrm{Im}\, \varphi| \,\Big|\, m!$$

So, if p is the smallest prime factor of $|G|$, then

$$m < p \Rightarrow \mathrm{Im}\, \varphi = 1 \quad \text{and} \quad m = p \Rightarrow |\mathrm{Im}\, \varphi| \in \{1, p\}.$$

In the case of the above theorem, the action is transitive, therefore $\mathrm{Im}\, \pi_H \neq 1$.
Therefore, $|\ker \pi_H| = |H|$ and $\ker \pi_H \leq H$, so proved.

Грummit. Sec 4.2.